Default Prediction Analysis

The analysis uses December 2014 borrower information to predict whether a borrower defaults during January-May 2015. After excluding borrowers with unknown income or gender, the sample contains approximately 1.74 million borrowers. The default rate is 6.13%.

Task 1: Default-prediction Table

Three models were estimated. The first includes a young-borrower indicator, gender, and a low-income indicator. The second adds bank credit limits and balances. The third adds JV nonbank issuer credit limits and balances. Young borrowers are defined as those below the 10th percentile of age, corresponding to borrowers under 31.

Young borrowers account for approximately 8.28% of the sample. This is less than 10% because borrowers aged exactly 31 are excluded under the strict “below” definition.

The results show that larger limits are associated with lower default risk, while larger balances are associated with higher risk, holding the other variables constant. The low-income association declines from +2.07 percentage points in the first model to +0.56 in the second and −0.20 in the third. Young borrowers have higher predicted default risk, with the estimated difference declining from 4.89 to 2.76 percentage points as credit information is added.

These results describe associations rather than causal effects. They do not establish that increasing a borrower’s limit would reduce their default risk.

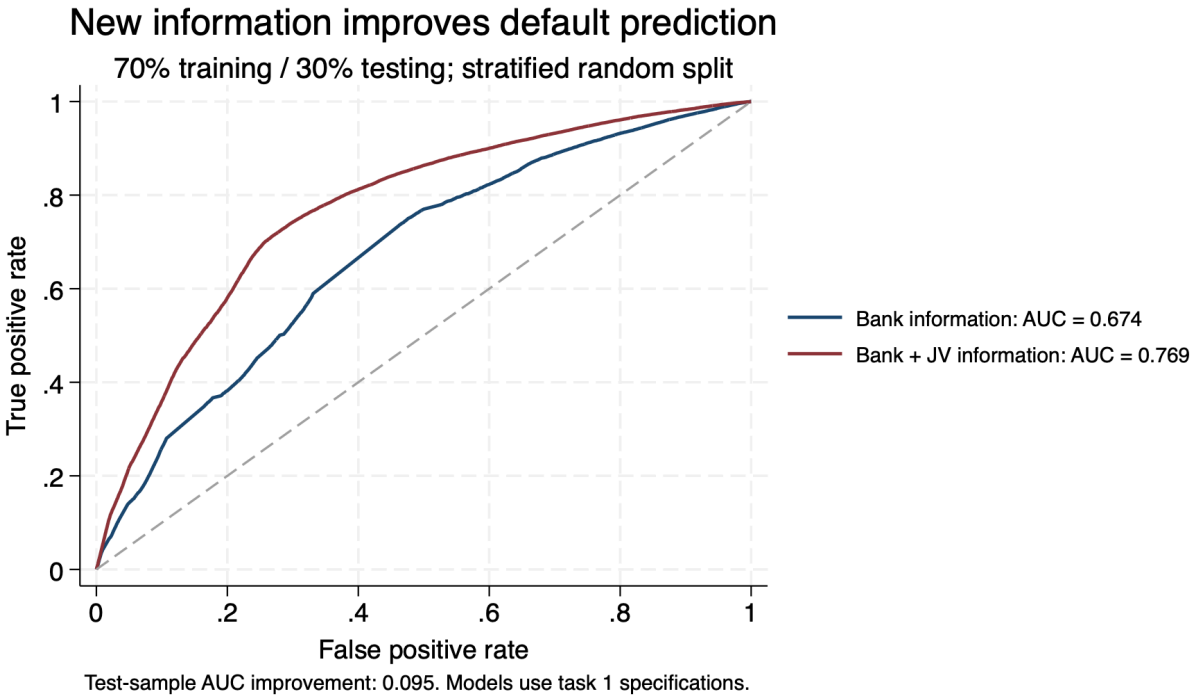
Predictor / statistic	(1) Demographics	(2) + Bank information	(3) + JV NBI information
Young (below age P10)	4.89*** (0.08)	4.50*** (0.08)	2.76*** (0.07)
Female	-0.93*** (0.04)	-0.96*** (0.04)	-0.98*** (0.04)
Low income (bracket 1)	2.07*** (0.04)	0.56*** (0.04)	-0.20*** (0.05)
Bank credit limit		-0.70*** (0.01)	-0.37*** (0.01)
Bank credit balance		1.28*** (0.01)	0.79*** (0.01)
JV NBI credit limit			-9.50*** (0.07)
JV NBI credit balance			11.98*** (0.08)
Default mean (%)	6.13	6.13	6.13
Observations	1,744,787	1,744,787	1,744,787
Pseudo R-squared	0.010	0.030	0.084

Task 2: Default Prediction using a 70/30 Split

70% of borrowers were used to estimate the models and the remaining 30% to evaluate their predictions. Both models were tested on the same borrowers.

Adding JV information increased test AUC from 0.674 to 0.769, an improvement of approximately 0.095. AUC measures how well a model ranks defaulters above nondefaulters. The increase shows that JV information helps distinguish between the two groups among borrowers who were not used to estimate the models.

The result supports the main finding in slide 27, although the sample and model specifications differ from the original analysis.

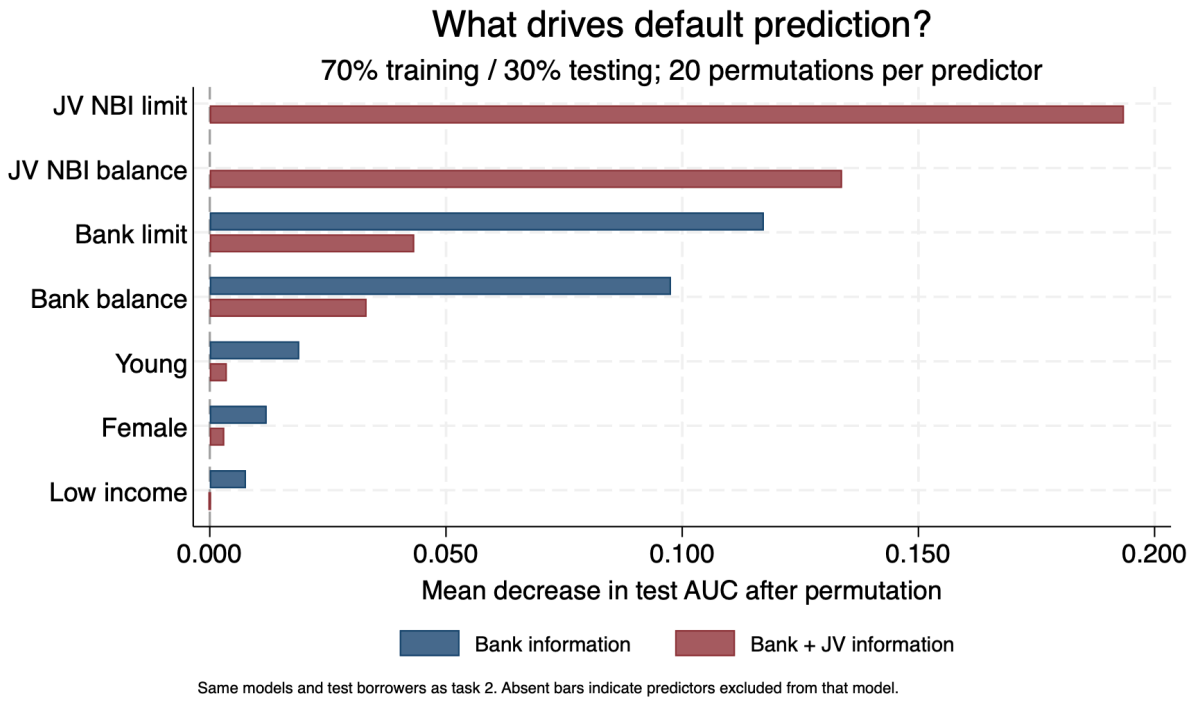


Task 3: Permutation Importance

Assessed each variable's importance by randomly shuffling its values among test borrowers and measuring how much AUC declined. This was repeated twenty times per variable.

JV credit limits and balances produced the largest declines in the expanded model. This indicates that the model relies heavily on these variables. Bank information became less important after JV information was added.

The young-borrower indicator made a modest contribution to prediction. Shuffling it reduced AUC by approximately 0.019 in the bank-only model and 0.004 in the expanded model. JV limits and balances remained substantially more important.



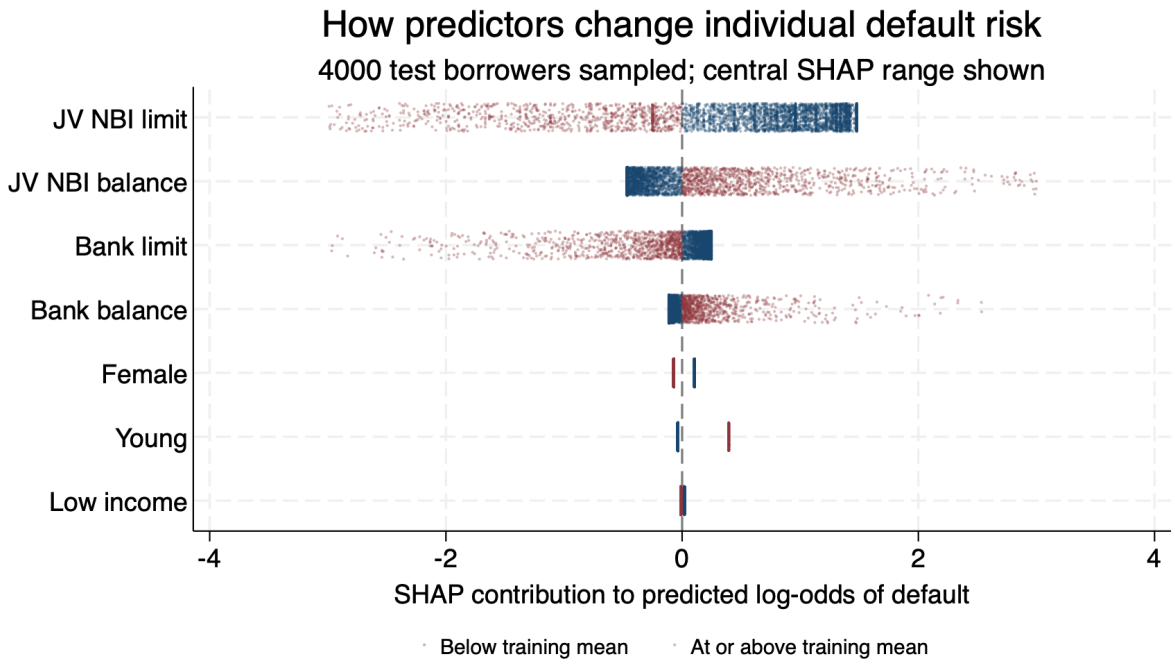
Same models and test borrowers as task 2. Absent bars indicate predictors excluded from that model.

Task 4: SHAP Values

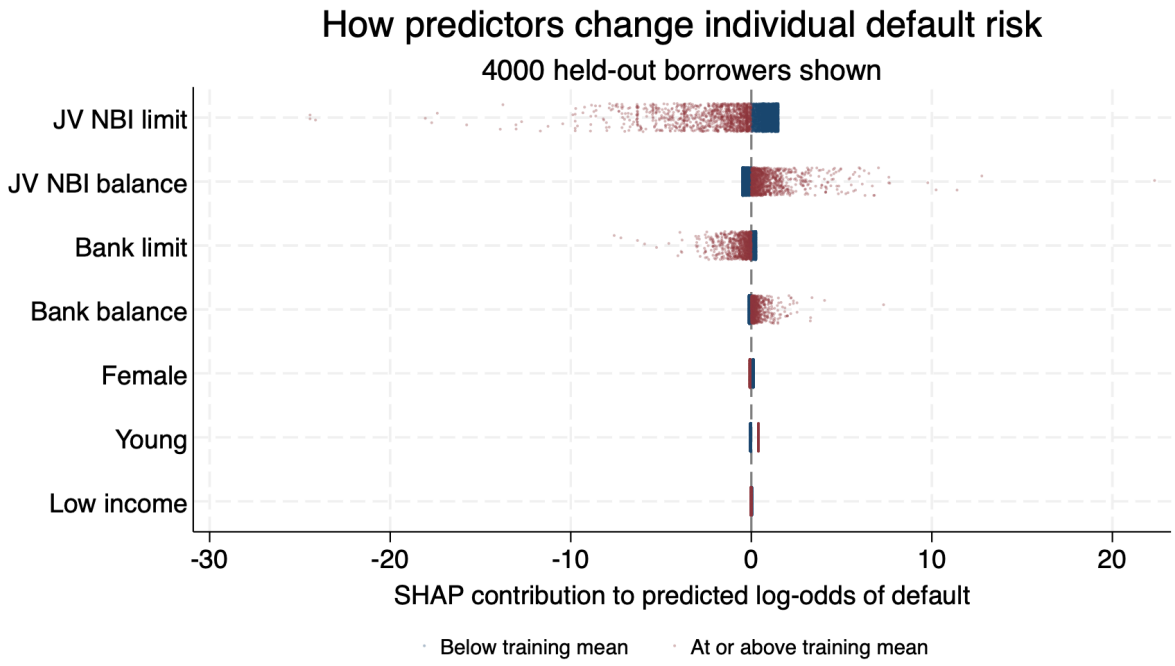
SHAP values were used to explain how individual borrower characteristics contribute to the expanded model's predictions.

Larger JV limits generally lower predicted risk relative to the training reference, while larger JV balances raise it. These variables also have the largest average contributions. Being young raises predicted risk relative to the training reference, but its average SHAP contribution is smaller than those of the credit variables. The contribution from low income remains small.

The plot shows the direction and size of these contributions for individual borrowers. Its values are measured in log-odds, rather than percentage-point changes in default probability.



421 extreme dots omitted here; full-range plot saved separately. Rows ranked using all test borrowers.



Right: higher risk. Left: lower risk. Rows ranked using all test borrowers.

Task 5: Proposed Additional Evaluation

The completed analysis shows that JV information improves risk ranking. Further evaluation would establish whether it also improves the accuracy of predicted probabilities and helps identify defaults at practical screening thresholds.

Average Precision and Precision-recall Curves

Since defaults are relatively uncommon, we should examine how effectively each model identifies defaulters without flagging too many nondefaulters. Average precision and precision-recall curves address this directly and complement the ROC analysis.

Log Loss and Brier Score

A model can rank borrowers well while assigning inaccurate default probabilities. Log loss and Brier score evaluate the probabilities themselves. Comparing these measures would show whether JV information improves probability estimates as well as ranking. Lower scores indicate better performance.

Calibration

Calibration compares predicted default probabilities with observed default rates. For example, among borrowers assigned approximately 10% risk, about 10% should default. This matters if the predictions will inform expected losses or lending decisions. Calibration plots would reveal whether either model systematically overstates or understates risk.

Performance at Practical Thresholds

Lenders typically make decisions at a particular screening threshold rather than across the entire ROC curve. We should compare how many defaulters each model identifies at the same false-positive rate. We should also examine the default rate among borrowers ranked in the highest-risk groups. These comparisons would make the improvement easier to interpret in a lending context.

Separate Contributions of JV Limits and Balances

We should estimate additional models that include JV limits alone, JV balances alone, and both together. This would establish how much improvement each variable adds when the model is allowed to adjust to the available information. It would complement permutation importance, which holds the fitted model fixed.

Grouped Permutation Importance

Credit limits and balances are related. Shuffling them separately can create unusual combinations. Shuffling them together would preserve their relationship and provide a measure of the model's reliance on the issuer's combined information.

Subgroup Performance

We should compare performance across age, gender, income, and existing bank exposure. This would show whether the improvement is broadly shared or concentrated among particular borrowers. Each comparison should report sample sizes and default rates to support interpretation.

Confidence Intervals

We should estimate confidence intervals for the difference between the models by repeatedly resampling test borrowers. Using the same resampled borrowers for both models would provide a fair comparison and indicate how precisely the improvement is estimated.

The recommended first step is to calculate average precision, log loss, Brier score, and calibration. We can then examine the separate contributions of JV limits and balances. These additional analyses have been proposed but have not yet been completed.